

UNIT 4 – MEMORY MANAGEMENT (DETAILED CLASS NOTES)

◇ 1. What is Memory Management?

Memory Management is a function of OS that:

- Allocates memory to processes
- Deallocates memory after use
- Keeps processes safe
- Uses memory efficiently

Real Example:

In your system:

- Chrome, VS Code, WhatsApp → all run together
- OS divides RAM between them

Engineering view:

Like CPU scheduling manages CPU, memory management manages RAM.

◇ 2. Basic Bare Machine

- No OS present
- Only one program runs
- Program gets full memory

Problems:

- No protection
- No multitasking
- CPU idle sometimes

Example:

- Early computers
- Embedded systems (simple controllers)

◇ 3. Resident Monitor

- Small OS stored in memory permanently
- Controls execution of programs

Functions:

- Load job
- Execute job

- Manage sequence

Limitation:

- Only one job at a time

◇ 4. Multiprogramming with Fixed Partitions

Memory is divided into fixed-size blocks.

Types:

1. Equal partitions
2. Unequal partitions

Working:

- One process per partition

Problem → Internal Fragmentation

Unused space inside partition

Example:

Partition = 200MB

Process = 120MB

→ 80MB wasted

◇ 5. Multiprogramming with Variable Partitions

Memory allocated dynamically.

Advantages:

- Better memory utilization

Problem → External Fragmentation

Free memory scattered in small pieces

Allocation Methods:

- First Fit → first available block
- Best Fit → smallest suitable block
- Worst Fit → largest block

Extra Topic: Compaction

- Combine small free blocks into one big block

◇ 6. Protection Schemes

Ensures processes do not interfere.

Techniques:

1. Base Register

Start address of process

2. Limit Register

Size of process

👉 Access allowed only within range

Real Example:

- **Linux / Windows / Android**
- Each app runs in separate memory space

◇ 7. Paging (Core Topic 🔥)

Memory divided into:

- Pages (logical)
- Frames (physical)

Address Translation:

Logical Address → Page Table → Physical Address

Advantages:

- No external fragmentation
- Efficient memory use

Disadvantage:

- Page table overhead

Extra Topic: Page Table

Stores mapping of pages to frames

◇ 8. Segmentation

Memory divided into logical segments:

- Code
- Data
- Stack

Address:

Segment number + offset

Advantage:

- Logical structure

Disadvantage:

- External fragmentation

◇ 9. Paged Segmentation

Combination of paging + segmentation

Working:

1. Divide into segments
2. Divide each segment into pages

Benefit:

- Structure + efficient allocation

◇ 10. Virtual Memory

Allows execution of large programs.

Key Idea:

- Only part of program in RAM
- Rest stored in disk

Benefits:

- Run large applications
- Better multitasking

Real Example:

- Windows shows "Virtual Memory / Page File"

◇ 11. Demand Paging

Pages are loaded only when required.

Steps:

1. Page requested
2. Not in memory → Page Fault
3. Load from disk

Benefit:

- Saves memory

◇ 12. Performance of Demand Paging

Important concept for numericals.

Terms:

- Memory Access Time (MAT)
- Page Fault Rate (p)

Insight:

- If page fault increases → performance decreases

◇ 13. Page Replacement Algorithms

When memory is full, replace pages.

◇ FIFO (First In First Out)

- Remove oldest page

👉 Simple but not efficient

◇ LRU (Least Recently Used)

- Remove least used page

👉 Used in real systems

◇ Optimal

- Remove page not needed in future

👉 Best but not practical

Extra Topic: Belady's Anomaly

- In FIFO, more frames can increase faults

◇ 14. Thrashing

System spends more time swapping pages than executing.

Cause:

- Too many processes
- High page fault rate

Effect:

- CPU utilization decreases
- System becomes slow

Solution:

- Reduce processes
- Increase RAM

◇ 15. Cache Memory Organization

Cache is small fast memory.

Purpose:

- Reduce memory access time

Types:

- L1 → fastest
- L2
- L3

Mapping Techniques:

- Direct Mapping
- Associative Mapping
- Set-Associative

◇ 16. Locality of Reference

Programs follow access patterns.

Types:**1. Temporal Locality**

Recently used data used again

2. Spatial Locality

Nearby data used

◇ 17. Related Important Topics (Extra for Exam)

◇ Internal vs External Fragmentation

Type	Where
Internal	Fixed partition
External	Variable partition

◇ Swapping

- Process moved between RAM and disk

Example:

Background apps in Android

◇ Working Set Model

- Minimum pages required for process

◇ Page Fault Types

- Minor → fast
- Major → slow (disk access)

Real OS Mapping

OS	Concept Used
Windows	Paging + Virtual Memory
Linux	Paging + Protection
Android	Demand Paging
Kali Linux	Secure memory isolation
macOS	Advanced memory + cache